

Firm and Industry Dynamics

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Firm and Industry Dynamics

- Many important problems in IO are inherently dynamic: firm growth, investment behavior, evolution of market leadership, dynamic responses to policy changes.
- Dynamic perspective can lead to new insights: high prices may be bad for current consumers but encourage investment, firms might rationally price below marginal cost to drive out rivals or move down learning curve, etc.
- Dynamic models get complex very fast; often require a reliance on numerical techniques and examples; models can be less transparent.
- Plan for lectures: (1) descriptive facts about firm and industry dynamics; (2) dynamic industry models; (3) models of firm behavior and empirics; (4) models of dynamic competition.

The Firm Size Distribution

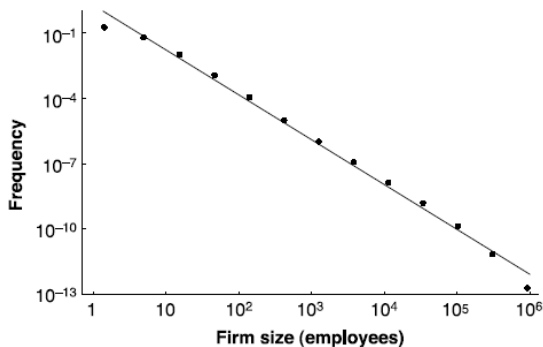
- Gibrat introduced systematic study of the firm size distribution.
- Main Finding: the firm size distribution is quite skewed and seems to follow certain regularities across time, countries, etc.
 - Gibrat argued that log-linear distribution fits the data well.
 - The Pareto distribution provides reasonable fit for the upper tail:

$$\Pr(S \geq s) = \left(\frac{s_0}{s}\right)^\alpha$$

where $s \geq s_0$ and $\alpha > 0$ ($\alpha = 1$ is the Zipf distribution).

Firm Size Distribution

- Upper tail of firm sizes, US Census, from Axtell (Science, 2001)



Firm Size Distribution

- Firm size distribution in Portugal: Cabral and Mata (AER, 2003)

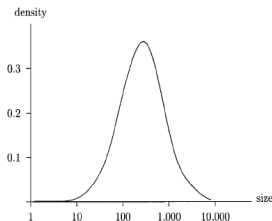


FIGURE 1. FIRM SIZE DISTRIBUTION, BASED ON EMPLOYMENT DATA FROM THE IF4 DATA SET

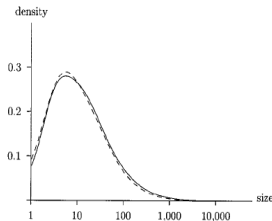


FIGURE 2. FIRM SIZE DISTRIBUTION IN 1983 (SOLID LINE) AND 1991 (DASHED LINE), BASED ON EMPLOYMENT DATA FROM THE *QUADROS DO PESSOAL* DATA SET

Firm Growth and Gibrat's Law

- Gibrat (1931) tried to explain log-normal size distribution with a model of firm growth in which a firm's growth each period was proportional to its current size.
- Gibrat's model:

$$x_t - x_{t-1} = \varepsilon_t y_{t-1}$$

so that, assuming $\log(1 + \varepsilon_t) \approx \varepsilon_t$, we have

$$\log x_t = \log x_0 + \varepsilon_1 + \varepsilon_2 + \dots + \varepsilon_t.$$

- Assume ε_t 's are $IN(\mu, \sigma^2)$. By CLT, x_t is log-normal as $t \rightarrow \infty$.
- Intuition: each period a new set of “opportunities” arise, and the probability of exploiting them is proportional to a firm's size.
- Analyses sometimes include entry: assuming the probability an opportunity is taken by a new entrant is constant over time.

Implications and Empirical Findings

- Limiting implications of the model (log-normal distribution of firm sizes) fit the data well, but...
- Test of firm growth rates consider

$$\log x_t = \alpha + \beta \log x_{t-1} + \zeta_t$$

and test if $\beta = 1$. Gibrat (1931) and later Simon (1958) found support for this using data on large public firms.

- More detailed Census data analyzed by Evans (1987), Dunne, Roberts and Samuelson, 1988, 1989) show departures.
 - Probability of survival increases with firm (or plant) size.
 - Proportional rate of growth of a firm (or plant) conditional on survival is decreasing in size.

Dunne, Roberts and Samuelson (1988, Rand J. Econ.)

TABLE 2 **Entry and Exit Variables for the U.S. Manufacturing Sector**
(Averages over 387 Four-Digit SIC Industries)

	1963-1967	1967-1972	1972-1977	1977-1982
Entry Rate (<i>ER</i>):				
All firms	.414	.516	.518	.517
Smallest firms deleted	.307	.427	.401	.408
Entrant Market Share (<i>ESH</i>):				
All firms	.139	.188	.146	.173
Smallest firms deleted	.136	.185	.142	.169
Entrant Relative Size (<i>ERS</i>):				
All firms	.271	.286	.205	.228
Smallest firms deleted	.369	.359	.280	.324
Exit Rate (<i>XR</i>):				
All firms	.417	.490	.450	.500
Smallest firms deleted	.308	.390	.338	.372
Exiter Market Share (<i>XSH</i>):				
All firms	.148	.195	.150	.178
Smallest firms deleted	.144	.191	.146	.173
Exiter Relative Size (<i>XRS</i>):				
All firms	.247	.271	.221	.226
Smallest firms deleted	.367	.367	.310	.344

Dunne, Roberts and Samuelson (1988, Rand J. Econ.)

TABLE 8 Market Shares, Average Firm Sizes, and Exit Rates of Entry Cohorts by Year (Means and Standard Deviations across 387 Industries)

	1963	1967	1972	1977	1982
<u>Market Shares</u>					
1963 Firms	1.00	.861 (.104)	.729 (.169)	.657 (.202)	.578 (.222)
1967 Entry Cohort		.139 (.104)	.083 (.062)	.067 (.054)	.053 (.044)
1972 Entry Cohort			.189 (.130)	.131 (.088)	.099 (.069)
1977 Entry Cohort				.147 (.109)	.098 (.074)
1982 Entry Cohort					.173 (.113)
<u>Average Size of Surviving Firms Relative to All Firms in the Industry</u>					
1963 Firms	1.00	1.49 (.406)	2.13 (1.13)	2.92 (1.90)	3.76 (3.37)
1967 Entry Cohort		.352 (.240)	.597 (.485)	.915 (.935)	1.32 (1.47)
1972 Entry Cohort			.396 (.250)	.686 (.455)	1.07 (.867)
1977 Entry Cohort				.308 (.202)	.560 (.357)
1982 Entry Cohort					.346 (.204)
<u>Cumulative Cohort Exit Rates</u>					
1963 Firms		.419 (.116)	.640 (.120)	.741 (.118)	.815 (.109)
1967 Entry Cohort			.639 (.100)	.790 (.075)	.876 (.063)
1972 Entry Cohort				.575 (.103)	.782 (.090)
1977 Entry Cohort					.632 (.111)

Descriptive Industry Dynamics (cont.)

- Entry and exit creates selection issue for empirical work: for example, we do not observe the growth rate of exiting firms, which are typically younger and smaller... More on this later.
- Gibrat's law does not provide a full theory
 - Entry and exit, mechanisms of growth aren't modelled
 - Still, strong regularity suggests a robust mechanism is at work.
- Further regularity ("turbulence"): Entry and exit are positively correlated across industries. This suggests differences in sunk costs across industries (cross-section) and in the variance of the process generating firm specific sources of change (which swamps common shocks to cost and demand conditions).

DRS (1988): Correlation between Entry and Exit

TABLE 7 Correlations between Industry Entry and Exit Variables (387 Four-Digit Industries)

	No Correction for Fixed Industry Effects				Correction for Fixed Industry Effects			
	1963-1967	1967-1972	1972-1977	1977-1982	1963-1967	1967-1972	1972-1977	1977-1982
	<u>Entry Rate</u>				<u>Entry Rate</u>			
<u>Exit Rate</u>								
1963-1967	.180	.363	.387	.323	-.249	.071	.123	-.005
1967-1972	.447	.274	.273	.363	.371	-.191	-.177	.118
1972-1977	.358	.408	.321	.328	.051	.137	-.129	-.081
1977-1982	.237	.324	.389	.304	-.114	-.029	.147	-.028
	<u>Entrant Market Share</u>				<u>Entrant Market Share</u>			
<u>Exiter Market Share</u>								
1963-1967	.741	.725	.743	.691	.308	-.116	-.037	-.167
1967-1972	.722	.770	.759	.703	.124	.154	-.058	-.228
1972-1977	.681	.800	.788	.784	-.153	.160	-.044	.032
1977-1982	.571	.691	.758	.804	-.287	-.172	.132	.354

Microeconomic Models of Size Distribution & Dynamics

- Lucas (1978) proposed early model of firm size distribution, based in the idea that firms leverage (heterogeneous) managerial talent.
 - Extended by Garicano (2000), Garicano & Rossi-Hansberg (2007)
- Jovanovic (1982) proposed a selection model for industry evolution.
 - Firms enter without knowing their true type, and learn about their type as they produce in the industry.
 - Over time less efficient types will realize that they are less efficient, produce less, and eventually exit.

Jovanovic (1982) Model

- Continuum of firms, each measure zero, so (non-stationary) price path is taken as given.
- Each entrant draw productivity θ from $N(\bar{\theta}, \sigma_\theta^2)$, and has costs $c(q)\xi(\theta + \epsilon_t)$ with ξ positive, bounded, strictly increasing.
- By observing cost a firm gains imperfect information about its type.
- Uncertainty about type is reduced over time:

$$N\left(\frac{\sigma^2 m + s^2 z}{\sigma^2 + s^2}, \frac{\sigma^2 s^2}{\sigma^2 + s^2}\right)$$

which can be written as:

$$N\left(\frac{hm + kz}{h + k}, h + k\right) \text{ with } h = \frac{1}{s^2} \text{ and } k = \frac{1}{\sigma^2}$$

where m is the prior and z is the observed realization.

- Each period firm chooses q to maximize expected profits

$$q_t p_t - c(q_t) E_{(t-1)} \xi(\theta + \epsilon_t)$$

- Solution has q decreasing in $E_{(t-1)} \xi(\theta + \epsilon_t)$, so firms that believe they're less efficient produce less.
- With a scrap value of exit W , the dynamic problem satisfies

$$V(x, n, t; p) = \pi(p_t, x) + \beta \int \max[W, V(z, n+1, t+1; p)] P(dz|x, n)$$

- There is a unique solution for V , and V is strictly increasing in x .
The net value of entry is $V(x_0, 0, t; p) - k$.

Predictions from Jovanovic Model

- 1 There is a threshold rule for exit decisions: firms exit if beliefs are below the cut-off, so exitors are small.
- 2 Beliefs about productivity (the x 's) tend to diverge over time, so firms become more heterogeneous, and the Gini coefficient (non-monotonically) increases as the industry matures.
- 3 Early on, there is more uncertainty about a firm's type, so the update is stronger and hence growth rates (for survivors) is greater. Later on, there is no update, and production stabilizes.
- 4 Thus, we have the regularity of younger and smaller firms growing faster, even if we correct for selection.

Related work on Firm Growth

- Sutton (1998): proposes “bounds” model of firm growth.
- Stanley et al. (1996), Axtell (2001), Rossi-Hansberg & Wright (2005), Gabaix (2009): theory and evidence related to power distribution of firm sizes.
- Industry life-cycle and “shake-out” models: add a further fact: at the beginning of an industry, there are often many firms, but later there may be relatively few.
 - Klepper (1993, and other papers with coauthors)
 - Jovanic and McDonald (1995)
- Also related are the Schumpeterian growth models: Aghion-Howitt (1992) and others.

Steady-State Industry Dynamics

- Steady-state models in which firms enter, grow and decline, and exit, but the overall distribution of firms remains stable. Dynamics created by random productivity shocks.
- Hopenhayn (1992) introduced basic model has perfect competition, extended to monopolistic competition and to international trade (two separate markets) by Melitz (2003).
- Later we will move to strategic interactions, which are somewhat more realistic and produce richer dynamic stories. On the other hand, they produce very weak comparative statics (almost anything can happen).
- Hopenhayn model allows us to characterize stationary distribution of firm size and other characteristics, and see how these change with changes in the parameters.

Hopenhayn (1992) Model

- Firm's productivity is denoted by φ , on $[0, 1]$.
- Assume $\varphi_{t+1} \sim F(\cdot | \varphi_t)$: higher φ_t means higher F by FOSD.
- Entrants draw their initial productivity from a distribution $G(\cdot)$.
- The timing of the model (within each period) is as follows:
 - Incumbents decide to stay or exit; entrants decide to enter or not..
 - Incumbent who stays pays c_f and gets a realization of its productivity, then produces.
 - Entrant who enters pays c_e as entry cost, get its productivity realization, and produces.
- The state of the industry at period t is denoted by μ_t , which is the measure over all productivity levels that are active.

Hopenhayn (1992) Model, cont.

- Firms have perfect foresight of output and input prices (p_t, w_t) .
- They solve the following program (note the difference in the timing from previous model):

$$v_t(\varphi, z) = \pi(\varphi, z_t) + \beta \max \left\{ 0, \int v_{t+1}(\varphi', z) F(d\varphi' | \varphi) \right\}$$

- Under regularity assumptions, the value function is increasing in φ , such that there is a cutoff point x_t below which firms exit. Similarly, entrants keep entering until the value from entry is zero, so that there is a positive mass M_t of entrants when

$$v_t^e(z) = \int v_t(\varphi', z) G(d\varphi') = c_e$$

- These imply an evolution process for the state of the industry:

$$\mu_{t+1}([0, \varphi']) = \int_{\varphi \geq x_t} F(\varphi' | \varphi) \mu_t(d\varphi) + M_{t+1} G(\varphi')$$

Hopenhayn (1992) Model: Results

- ① There exists a stationary competitive equilibrium of the industry.
- ② There exists $c^* > 0$ such that for any $c_e < c^*$ there exists a competitive stationary equilibrium with positive entry and exit.
- ③ The equilibrium is unique under certain (mild) assumptions.
- ④ Comparative statics:
 - Size distribution of firms increases with age
 - The same goes for profits and value distribution
 - Increase in entry cost reduces entry (M) and turnover ($M/\mu(s)$).
- ⑤ (Melitz) If firms must make decisions that involve fixed cost and variable benefits that depend on size, largest and most productive firms will invest (example: exporting).

Empirical Application: Olley-Pakes (1996)

- Attempt to deal with selection in production function estimation.
- Concern: older and bigger firms are less likely to exit even when they get a negative productivity shock. If we do not account for selection, we may get biased coefficients on age and capital.
- An example for attrition bias:

$$E(y_t | y_{t-1}, \chi_t = 1) = \theta_0 + \theta_1 y_{t-1} + E(\omega_t | y_{t-1}, \chi_t = 1)$$

To think about the bias, we need a model for attrition. If attrition depends on a cutoff strategy (i.e. $\chi_t = 1$ iff $y_t > \underline{y}$) then higher y_{t-1} 's will survive with lower ω_t 's and θ_1 will be biased downwards.

- Production function estimation and the implication of the telecom industry deregulation:

$$y_{it} = \beta_0 + \beta_l l_{it} + \beta_a a_{it} + \beta_k k_{it} + \omega_{it} + \eta_{it}$$

- Problems:
 - endogeneity of labor (β_l biased upwards)
 - selection (β_k biased downwards).
- Suppose exit decisions follow a threshold rule: remain in the market iff $\omega_{it} > \underline{\omega}(a_{it}, k_{it})$, where $\underline{\omega}$ is decreasing in k .
- That is, bigger firms less likely to exit, as suggested by the descriptive statistics. (Note the “large market” assumption: exit decision is independent of other firms,)
- Approach: recover persistent productivity ω_{it} from observed investment: if investment $i(\omega_{it}, a_{it}, k_{it})$ is strictly monotone in ω_{it} , we can invert to find $\omega_{it} = h(i_{it}, a_{it}, k_{it})$.

Olley and Pakes Estimation Procedure

- Step 1:

$$y_{it} = \beta_l l_{it} + \phi(i_{it}, a_{it}, k_{it}) + \eta_{it}$$

where

$$\phi(i_{it}, a_{it}, k_{it}) = \beta_0 + \beta_a a_{it} + \beta_k k_{it} + h(i_{it}, a_{it}, k_{it})$$

and is estimated nonparametrically.

- Step 2: estimate survival probabilities by a probit of survival on (i_{it}, a_{it}, k_{it}) nonparametrically. Write survival probabilities as $P(i_{it}, a_{it}, k_{it})$.

Olley and Pakes Estimation Procedure

- Step 3:

$$\begin{aligned} y_{it+1} - \beta_l l_{it+1} &= \beta_a a_{it+1} + \beta_k k_{it+1} \\ &\quad + g(P_{it}(i_{it}, a_{it}, k_{it}), \phi(i_{it}, a_{it}, k_{it}) \\ &\quad - \beta_a a_{it} - \beta_k k_{it}) + \eta_{it} \end{aligned}$$

where

$$\begin{aligned} &g(P_{it}(i_{it}, a_{it}, k_{it}), \phi(i_{it}, a_{it}, k_{it}) - \beta_a a_{it} - \beta_k k_{it}) \\ &= E(\omega_{it+1} | survival) = E(\omega_{it+1} | \omega_{it}, \omega_{it+1} > \underline{\omega}(a_{it}, k_{it})) \end{aligned}$$

which is some function of P_{it} and

$$\omega_{it} = \phi(i_{it}, a_{it}, k_{it}) - \beta_a a_{it} - \beta_k k_{it}$$

under reasonable assumptions about $F(\omega_{it+1} | \omega_{it})$.

TABLE I
CHARACTERISTICS OF THE DATA

Year	Plants	Firms	Shipments (billions 1982 \$)	Employment
1963	133	104	5.865	136899
1967	164	131	8.179	162402
1972	302	240	11.173	192248
1977	405	333	13.468	192259
1982	473	375	20.319	222058
1987	584	481	22.413	184178

TABLE II
BELL COMPANY EQUIPMENT PROCUREMENT
(PERCENT PURCHASED FROM WESTERN ELECTRIC)

1982	1983	1984	1985	1986 ^E
92.0	80.0	71.8	64.2	57.6

^E Estimated for 1986.

Source: NTIA (1988, p. 336, and discussion pp. 335–337).

TABLE III
ENTRANTS ACTIVE IN 1987

	Number	Share of Number Active in 1987 (%)	Share of 1987 Shipments (%)	Share of 1987 Employment (%)
Plants: New since 1972	463	79.0	32.8	36.0
Firms: New since 1972	419	87.0	30.0	41.4
Plants: New since 1982	306	52.0	12.0	13.5
Firms: New since 1982	299	60.1	19.4	27.5

Olley and Pakes Application

TABLE VI
ALTERNATIVE ESTIMATES OF PRODUCTION FUNCTION PARAMETERS^a
(STANDARD ERRORS IN PARENTHESES)

Sample:	Balanced Panel		Full Sample ^{c, d}						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	Nonparametric F_ω	
Estimation Procedure	Total	Within	Total	Within	OLS	Only P	Only h	Series	Kernel
Labor	.851 (.039)	.728 (.049)	.693 (.019)	.629 (.026)	.628 (.020)			.608 (.027)	
Capital	.173 (.034)	.067 (.049)	.304 (.018)	.150 (.026)	.219 (.018)	.355 (.02)	.339 (.03)	.342 (.035)	.355 (.058)
Age	.002 (.003)	-.006 (.016)	-.0046 (.0026)	-.008 (.017)	-.001 (.002)	-.003 (.002)	.000 (.004)	-.001 (.004)	.010 (.013)
Time	.024 (.006)	.042 (.017)	.016 (.004)	.026 (.017)	.012 (.004)	.034 (.005)	.011 (.01)	.044 (.019)	.020 (.046)
Investment	—	—	—	—	.13 (.01)	—	—	—	—
Other Variables	—	—	—	—	—	Powers of P	Powers of h	Full Polynomial in P and h	Kernel in P and h
# Obs. ^b	896	896	2592	2592	2592	1758	1758	1758	1758

TABLE IX
INDUSTRY PRODUCTIVITY GROWTH RATES^a

Time Period	(1) Full Sample	(2) Balanced Panel
1974–1975	–.279	–.174
1975–1977	.020	–.015
1978–1980	.146	.102
1981–1983	–.087	–.038
1984–1987	.041	.069
1974–1987	.008	.020
1975–1987	.032	.036
1978–1987	.034	.047

^aThe numbers in Table IX are annual averages over the various subperiods.

TABLE X
 VARIABLE COST EFFICIENCY^a
 (MINIMUM COST OF PRODUCTION DIVIDED BY ACTUAL COST OF PRODUCTION)

Years	Total	Interfirm	Intrafirm
1974–1977	.77	.84	.91
1978–1980	.69	.76	.91
1981–1983	.65	.72	.91
1984–1987	.72	.80	.89

Olley and Pakes: Summary

- Main identification assumption: (a) productivity depends on only one unobservable dimension; and (b) the investment equation is invertible in productivity.
- This would not be the case if either of the following: 1. investment is not strictly monotone (e.g. fixed costs: see Levinsohn and Petrin, 2003); 2. not all the productivity effect is transmitted to the investment equation; 3. there are other unobservable factors which affect investment but do not enter the production function (e.g. interest rate fluctuations). (If interested, see a recent paper by Akerberg, Caves, and Frazer).
- Check also Griliches and Mairesse (NBER Working Paper 5067, 1995) for a more general review.

Single Agent Dynamics: Background Theory

- Consider a firm with time t payoffs given by:

$$f(k_t, i_t, z_t) = \pi(k_t, z_t) - c(i_t)$$

where k is capital, z is the price of the output, and i is investment.

- Assume that $\pi(k, z)$ is increasing in both its arguments, and that z follows an exogenous Markov process (i.e. $z_{t+1} \sim F(\cdot | z_t)$, where $F(\cdot | z)$ is stochastically increasing.
- Firm will be assumed to have partial control over the evolution of capital, so that $k_{t+1} \sim P(k_t, i_t)$ and that $P(k, i)$ is stochastically increasing in both k and i . The fact that it is increasing in i and that profits are increasing in k provides the reason to invest.
- A special (degenerate) case we get $k_{t+1} = (1 - \delta)k_t + i_t$. On the other hand, this formulation allows for other stuff, such as research and development, advertising, and exploration.

Investment Dynamics, cont.

- Firm's problem is whether to exit ($\chi = 0$) or not ($\chi = 1$), and if he stays in to decide on the investment level.
- If the monopolist exits he obtains a scrap value of ϕ and cannot reenter the market again.
- Under standard regularity conditions (in particular, bounded profit function) the expected discounted value of future profits satisfies the Bellman equation

$$V(k, z) = \max \left\{ \phi, \max_{i \geq 0} f(k, i, z) + \beta \int V(k', z') dP_k(k'|k, i) dP_z(z'|z) \right\}$$

- To actually “solve” the model, we need to find $V(\cdot)$, and the policy functions $\chi(\cdot)$, and $i(\cdot)$, all of which depend on the state variables k, z .

Contraction Mapping Theorem

- Define T to be an operator that maps functions to functions s.t.

$$Tv(k, z) = \max \left\{ \phi, \max_{i \geq 0} f(k, i, z) + \beta \int v(k', z') dP_k(k'|k, i) dP_z(z'|z) \right\}$$

- Finding a solution to the Bellman equation is equivalent to finding a fixed point of T , i.e. a V such that $TV = V$ (at every point k, z).
- Let (S, ρ) be a complete metric space. The function $T : S \rightarrow S$ is a contraction mapping (with modulus β) if for some $\beta \in (0, 1)$, $\rho(Tf, Tg) \leq \beta \rho(f, g)$ for all $f, g \in S$.
- In applications, often have S as the set of bounded functions on R^I (or a subset of it) and ρ the sup norm, i.e. $\rho(f, g) = \sup_{x \in R^I} |f(x) - g(x)|$.

Contraction Mapping Theorem

- *Theorem: (Contraction Mapping)* If (S, ρ) is a complete metric space and $T : S \rightarrow S$ is a contraction mapping with modulus β then:
 - ① T has exactly one fixed point, denoted by $V \in S$
 - ② For any $v_0 \in S$, $\rho(T^n v_0, V) \leq \beta^n \rho(v_0, V)$, and hence by completeness $\lim_{n \rightarrow \infty} T^n v_0 = V$
 - ③ $\rho(T^n v_0, V) \leq (1 - \beta)^{-1} \rho(v_n, v_{n-1})$.
- Part (1) says there is a unique value function for the problem; (2) provides an iterative way to compute it; and (3) gives an upper bound on the computation error.
- We can also derive monotonicity properties. Suppose that if v is increasing, then so is Tv . Then V must also be increasing (in k for example).

Examples of Dynamic Decision Problems

- Optimal stopping problems, i.e. binary decisions...
 - Patent renewal and expiration (Pakes, 1986)
 - Equipment replacement (Rust, 1987)
 - Sequential search (many, e.g. Hortacsu et al. 2009)
 - Loan repayment and default (Jenkins, 2008)
- Multiple or continuous choice problems
 - Capital investment decisions (e.g. sS models).
 - Pricing decisions with unknown demand (experimentation)
 - Pricing decisions with fixed inventory (“revenue management”)
 - Production decisions with learning by doing.
 - Discrete product choice with switching costs.

Pakes (1986): Optimal Stopping Model of Patent Renewal

- There is a big literature on patents, because this is one of the few measurable outcomes of research and development or innovative activity.
- Pakes' paper is about estimating the private returns to holding a patent – a very hard but important problem. He uses data on patent renewals in three European countries (France, Germany, UK).
- The key idea is that the annual renewal of a patent has two types of benefits: the returns during the coming year and the option to renew it later on. If the patent is not renewed then the assignee loses the rights forever.

- The renewal decision is modelled as the solution to a finite-horizon dynamic programming problem:

$$V(a, r) = \max \left\{ 0, r - c_a + \beta \int V(a + 1, r') dF(r'|r) \right\}$$

- c_a is the renewal fee (increasing in a)
- r is the current returns on holding the patent, which is known and follows a Markov process with natural monotonicity.
- The problem is solved backwards, with the last period solution is simply $V(L, r) = \max \{0, r - c_L\}$.
- This gives a cutoff r_a^* for renewal at each age. The assumptions above guarantee that r_a^* is increasing in a (the option value is decreasing over time). This shows up in Figure 1.

Pakes (1986): Optimal Renewal Policy

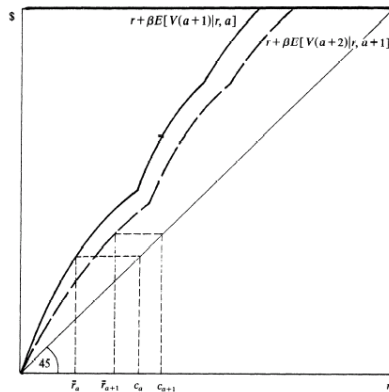


FIGURE 1.—Determining $\{\bar{r}_a\}_{a=1}^L$

- Pakes uses parametric assumptions on patent value:

$$r_{a+1} = \begin{cases} 0 & \text{with prob. } \exp(-\theta r_a) \\ \max\{\delta r_a, z\} & \text{with prob. } 1 - \exp(-\theta r_a) \end{cases} w$$

- Assume density of z is a two-parameter exponential:

$$q_a(z) = \frac{1}{\phi^{a-1}\sigma} \exp\left(-\frac{\gamma + z}{\phi^{a-1}\sigma}\right).$$

- Also assumes initial returns are lognormal

$$\log r_1 \sim N(\mu, \sigma_R).$$

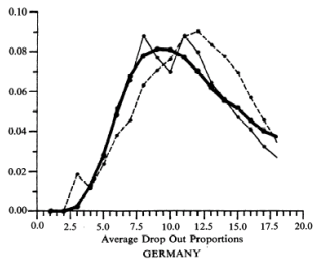
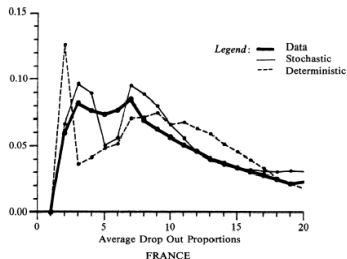
- Pakes tries to estimate parameters of the return distribution (the initial distribution, and the Markov process).
- How? He observes the fraction of patents that are not renewed each year, and wants the model to match the distribution of dropout times.
- Denote the predicted dropout probability by

$$\begin{aligned}\pi(a) = & \Pr \{r_{a-1} > r_{a-1}^*, \dots, r_1 > r_1^*\} - \\ & - \Pr \{r_a > r_a^*, r_{a-1} > r_{a-1}^*, \dots, r_1 > r_1^*\}\end{aligned}$$

and use maximum likelihood to estimate this.

- Note that while it is not hard to solve for the value function, there is no way to get these distributions analytically. Pakes uses simulation: given a set of parameters he simulates many patents, and has them go through the dropout process. Then he calculate the dropout frequencies, and uses these to estimate.

Pakes (1986), cont.



- Paper was of the first uses of simulation for estimation in economics (also in Lerman and Manski, 1981).
- One limitation of the paper is that renewing a patent is cheap, but the aggregate value of patents is really driven by the upper tail of values (most patents turn out to be worth zero).
- This upper tail is identified only by the strong functional form assumption – but not obvious how else you might pin it down.

Rust (1987): Capital Replacement Decisions

- The paper is about Harold Zurcher's bus engine replacement decisions, but the point is really the method rather than bus maintenance in Madison, Wisconsin.
- The data in hand is monthly observations on engine mileage and whether the engine was replaced or not.
 - 162 buses belonging to Madison Metro
 - Observed from Dec. 1974 to May 1985.
 - Bus engines typically replaced every five years with around 200,000 miles.
 - Eight different types of buses – different makes and models.

TABLE IIa
SUMMARY OF REPLACEMENT DATA
(Subsample of buses for which at least 1 replacement occurred)

Bus Group	Mileage at Replacement				Elapsed Time (Months)				Number of Observations
	Max	Min	Mean	Standard Deviation	Max	Min	Mean	Standard Deviation	
1	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0
3	273,400	124,800	199,733	37,459	74	38	59.1	10.9	27
4	387,300	121,300	257,336	65,477	116	28	73.7	23.3	33
5	322,500	118,000	245,291	60,258	127	31	85.4	29.7	11
6	237,200	82,400	150,786	61,007	127	49	74.7	35.2	7
7	331,800	121,000	208,963	48,981	104	41	68.3	16.9	27
8	297,500	132,000	186,700	43,956	104	36	58.4	22.2	19
Full Sample	387,400	83,400	216,354	60,475	127	28	68.1	22.4	124

Rust (1987), cont.

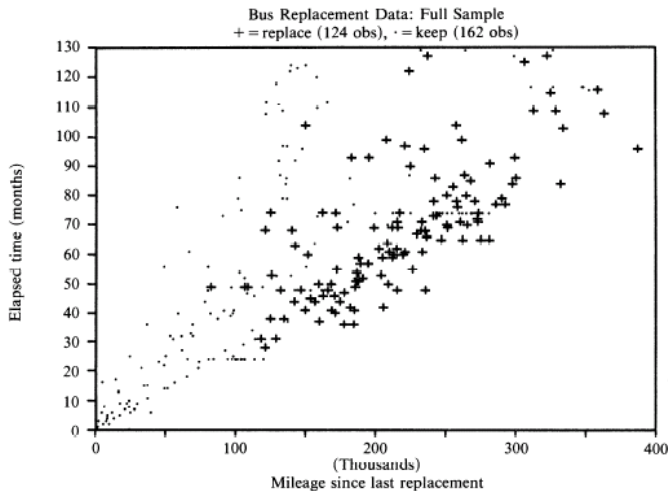


FIGURE 1

Rust (1987): Capital Replacement Decisions

- Rust models the replacement decision as a regenerative optimal stopping problem, where the state variable is the mileage x , and operating costs are $c(x, \theta)$.
- The value function is:

$$V(x) = \max \left\{ \begin{array}{l} -\Delta P - c(0) + \beta \int V(x') dF(x'|0), \\ -c(x) + \beta \int V(x') dF(x'|x) \end{array} \right\}$$

where ΔP is the additional cost from replacement, and F governs how many miles are driven.

- Under certain assumptions the optimal policy is a threshold policy: replace if $x \geq \bar{x}$.
- Problem: the model predicts deterministic replacement, but we have a wide range of x 's for the engines that are being replaced, so we must have more degrees of freedom to the model.

- Rust points out that with the structural model, it is internally inconsistent to just add a noise to the decision.
- Instead, Rust adds a structural error term into the replacement price: so the optimal replacement policy is “replace” if $\varepsilon \leq h(x; \theta)$ with h increasing in x .
- We estimate $F(\cdot)$ and $c(\cdot)$. The way to do so is to recalculate the value function (it is a contraction) for each new set of the parameters, and then recompute the likelihood $\Pr(i_t | x_t) \Pr(x_t | x_{t-1})$.
- This is the “nested fixed point” procedure. We’ll talk more about this, and alternative approaches, next time.

Benkard (2000): Learning in Aircraft Production

- Classic observation about aircraft production (e.g. Wright, 1936; Alchian, 1963): unit costs decrease with cumulative production.
- Benkard estimates learning curves for the Lockheed L-1011 TriStar — shows that cost efficiencies can be lost if there are gaps in production (forgetting).
- Data: labor (man-hour) requirements for 250 L-1011s produced between 1970 and 1984.

Benkard (2000): Learning in Aircraft Production

VOL. 90 NO. 4

BENKARD: DYNAMICS OF LEARNING AND FORGETTING

1039

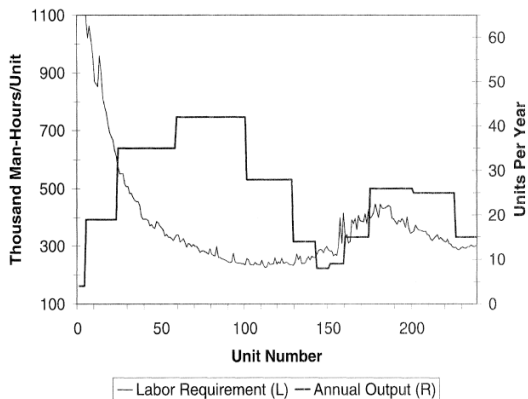


FIGURE 1. L-1011 PRODUCTION: DIRECT LABOR REQUIREMENT AND YEARLY OUTPUT

Benkard (2000): Learning in Aircraft Production

- Basic learning model:

$$\ln L_i = \ln A + \theta \ln E_i + \gamma \ln S_i + \varepsilon_i$$

where L_i is labor input per unit, A a constant, E_i is experience, and S_i is the line speed or current production rate.

- Modeling experience (not cumulative past output, but assume $E_1 = 1$).

$$E_t = \delta E_{t-1} + q_{t-1}$$

Also extends to allow for partial learning from related but non-identical models.

- Problem: ε_i not iid and may not be independent of E_i, S_i if productivity is persistent and affects production rate.
- Instrument for production rate with demand shifters (world & OECD GDP, price of oil), and cost shifters (aluminum price, US manufacturing wages, plus lags).

Benkard (2000): Learning in Aircraft Production

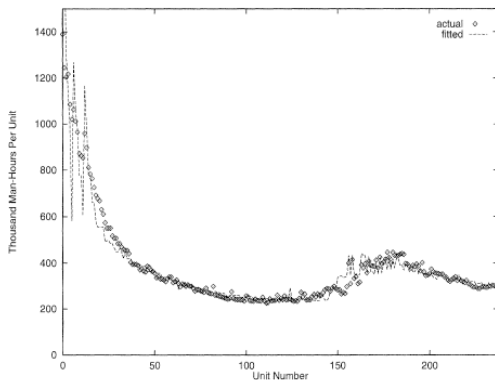


FIGURE 4. ORGANIZATIONAL FORGETTING AND INCOMPLETE SPILLOVERS (REGRESSION 10)

Strategic Models: Markov Perfect Equilibrium

- Dynamic models get messier with multiple agents. Standard solution concepts for strategic games need not give sharp predictions (recall the folk theorems for infinitely repeated games), so we look for refinements.
- The most tractable refinement for empirical work is Markov Perfect Equilibrium: players use strategies that depend on a common set of directly payoff-relevant state variables.
- An important early contribution, by Maskin and Tirole, showed that the concept of MPE together with alternating moves can formalize several IO stories that were in the literature. One is kinked demand curve and Edgeworth cycles, and the other, that we do in class, is some sort of a contestable market.
- More recent literature, initiated by Ericson and Pakes, studies a broader class of dynamic oligopoly models using MPE and attacks the problem from a computational standpoint.

Maskin-Tirole (1986): Duopoly with Large Fixed Costs

- Industry demand: $p = 1 - Q$, and $f = \frac{1}{8}$.
- So $\pi_t^1(q_t^1, q_t^2) = (1 - q_t^1 - q_t^2)q_t^1 - \frac{1}{8}$ if $q_t^1 > 0$ and zero otherwise.
- Firms maximize long-run profits with discount factor δ .
- Player 1 moves in odd periods, player 2 moves in even periods.
- Decisions involve commitment: they last for at least one more period.
- Consider the static game: best response conditional if opponent doesn't produce is to produce $1/2$, with optimal production falling linearly in q_j , down to zero if $q_j \geq 1 - 2^{-1/2} \approx 0.3$.
 - Two pure equilibrium: $(q^m, 0)$ and $(0, q^m)$
 - One mixed equilibrium that is symmetric.
- None of these equilibrium reflects a “contestable” outcome.
- Dynamic model has many, many SPE...

Maskin-Tirole (1986): Duopoly with Large Fixed Costs

- Using MPE in which strategies depend only on the opponent's q (but not on past choices!) we obtain the following unique symmetric equilibrium (for high enough values of δ):

$$s(q) = \begin{cases} \bar{q} & \text{if } q < \bar{q} \\ 0 & \text{if } q \geq \bar{q} \end{cases}$$

where $\bar{q}$ solves $\pi(\bar{q}, \bar{q}) + \frac{\delta}{1-\delta} \pi(\bar{q}, 0) = 0$.

- So as δ goes to 1 we have that $\bar{q}$ approaches the competitive level.
- The dynamic model rationalizes a contestable outcome!

- Define the two continuation values for each player:

$$V^1(a^2) = \max_{a^1} [\pi^1(a^1, a^2) + \delta W^1(a^1)]$$

and

$$W^1(a^1) = \pi^1(a^1, s^2(a^1)) + \delta V^1(s^2(a^1))$$

- We can substitute to get the Bellman equation:

$$V^1(a^2) = \max_{a^1} [\pi^1(a^1, a^2) + \delta \pi^1(a^1, s^2(a^1)) + \delta^2 V^1(s^2(a^1))]$$

- The MPE is a pair of reaction functions that are best response to each other.

Proof of Maskin-Tirole

- *Lemma 1:* $s(q)$ is non-increasing.

Proof: A standard comparative static w/ π submodular.

- *Lemma 2:* $s(q) > 0 \Rightarrow s(q) > q$.

Proof: Suppose not, then $s(q) \leq q$ and $s(s(q)) \geq s(q)$ (by Lemma 1). So $s^3(q) \leq s^2(q)$, etc. But then one of the firms always produces less, and hence is losing money with certainty (fixed costs are too high to support both of them).

- *Lemma 3:* $\exists \bar{q}$ s.t. $\forall q > \bar{q}: s(q) = 0$, $\forall q < \bar{q}: s(q) > q > 0$

Proof: The first part is just driven by the fact that for q large enough $s(q) = 0$ (may need marginal cost to have this). Then we can set $\bar{q} = \inf\{q : s(q) = 0\}$, and apply Lemmas 1 and 2.

Proof of Maskin-Tirole

- *Lemma 4:* If $q > \bar{q}$ then the firm reacting to q stays out forever.

Proof: Suppose not, then it sets $r = s(q) > q$ (Lemma 3). We also know that q was set as a response to r' , namely $q = s(r') > r'$ (Lemma 3). So $s(q) > s(r')$ for $q > r'$, which contradicts Lemma 1.

- *Lemma 5:* For δ close to 1, $\bar{q} > q_m = 1/2$.

Proof: Suppose not, then $s(q_m + \varepsilon) = 0$, but then $\pi(q_m + \varepsilon, q_m + \varepsilon) + \frac{\delta}{1-\delta} \pi(q_m + \varepsilon, 0) > 0$ so entry is not deterred, which is a contradiction.

- *Lemma 6:* $s(q) = \bar{q}$ if $q \leq \bar{q}$, zero otherwise.

Proof: Suppose $q < \bar{q}$, we know that $s(q) > q > 0$ and $s(s(q)) = 0$. But then it is optimal to set $s(q) = \bar{q}$ because $\bar{q} > q_m$. For any $q < \bar{q}$ we have that $s(q) = \bar{q}$, but s is non-increasing so $s(\bar{q}) \leq \bar{q}$ and hence must be zero by Lemma 3.

- *Lemma 7:* q is the greater root of $\pi(\bar{q}, \bar{q}) + \frac{\delta}{1-\delta} \pi(\bar{q}, 0) = 0$.
Proof: If it is strictly less than zero, then by setting $\bar{q}$ we lose money against $\bar{q} - \varepsilon$. If it is strictly more then we can earn even against $\bar{q} + \varepsilon$. If it is not the greater root we can do better by setting higher quantity (concave function).
- **Summary:** So, we show the idea of contestability in a dynamic framework. The key is the Markov structure with small state space, and the alternating move assumption, which is simplifying but not as crucial. Related ideas are obtained for kinked demand curve etc.

Empirical Model of Dynamic Competition

- Firms, $i = 1, \dots, N$
- Time $t = 1, \dots, \infty$
- State at time t , $s_t \in S \subset \mathbb{R}^L$, commonly observed.
- Actions at time t : $a_t = (a_{1t}, \dots, a_{Nt}) \in A$.
- Private “shocks” $v_{it} \in \mathcal{V}_i \subset \mathbb{R}^M$, drawn iid from $G(\bullet | s_t)$.
- Payoff functions, $\pi_i(a_t, s_t, v_{it})$
- Discount factor $\beta < 1$.
- Expected future profits from t on:

$$\mathbb{E} \left[\sum_{\tau=t}^{\infty} \beta^{\tau-t} \pi_i(a_{\tau}, s_{\tau}, v_{it}) \middle| s_t \right]$$

- State transitions: s_{t+1} drawn from $P(\bullet | a_t, s_t)$.

Strategies, Value Functions and Equilibrium

- Markovian strategy for firm i : $\sigma_i : S \times \mathcal{V}_i \rightarrow A_i$
- Strategy profile: $\sigma = (\sigma_1, \dots, \sigma_N) : S \times \mathcal{V} \rightarrow A$.
- Value functions:

$$V_i(s|\sigma) = \mathbb{E}_v \left[\pi(\sigma(s, v), s, v_i) + \beta \int V_i(s'|\sigma) dP(s'|\sigma(s, v), s) \middle| s_t \right].$$

- Definition: A strategy profile σ is a *Markov Perfect Equilibrium* if for every firm i , σ_i is a best response to σ_{-i} . That is, for every firm i , state s and Markov strategy σ'_i

$$\begin{aligned} V_i(s|\sigma) &\geq V_i(s|\sigma'_i, \sigma_{-i}) \\ &= \mathbb{E}_v \left[\begin{array}{c} \pi(\sigma'_i(s, v_i), \sigma_{-i}(s, v_{-i}), s, v_i) \\ + \beta \int V_i(s'|\sigma'_i, \sigma_{-i}) dP(s'|\sigma_i(s, v_i), \sigma_{-i}(s, v_{-i}), s) \end{array} \middle| s_t \right]. \end{aligned}$$

Example: Ericson-Pakes Dynamic Oligopoly

- N firms, each either an incumbent or a potential entrant.
 - If firm exits, will be replaced by a new potential entrant — no re-entry.
- Each firm has a “productivity state” $z_{it} \in \{0, 1, 2, \dots, Z\}$, where 0 means firm is not in the market.
- Overall state $s_t = (z_{1t}, \dots, z_{It}, m_t)$, where m_t is state of demand.
- Incumbent firm actions: $a_{it} = (p_{it}, l_{it}, \chi_{it})$, price, investment, in/out.
- Potential entrant actions: $a_{it} = \chi_{it}$ (only in/out).
- Demand: $q_{it} = Q_i(s_t, p_t)$

Example: Ericson-Pakes (1995) Dynamic Oligopoly

- Incumbent firm profits at time t :

$$\pi_i(a_t, s_t, v_i) = \underbrace{q_{it}(p_{it} - mc(q_{it}))}_{\text{static profits}} - \underbrace{C(l_{it}, v_{it}^l) + (1 - \chi_{it})\phi}_{\text{cost/benefit of dynamic choices}}$$

where v_{it}^l is a private idiosyncratic shock to investment cost; ϕ is a scrap value.

- Potential entrant profits at time t :

$$\pi_i(a_t, s_t, v_i^e) = \chi_{it} v_i^e$$

where v_{it}^e is a private idiosyncratic shock to the cost of entering the market.

- State transitions:

- m_t evolves exogenously: m_{t+1} drawn from $P^m(\bullet | m_t)$.
- z_t evolves endogenously: $z_{i,t+1}$ drawn from $P^z(\bullet | z_{it}, \chi_{it}, l_{it})$

Example: Ericson-Pakes Dynamic Oligopoly, continued

- Markovian Policies: at time t , σ_{it} specifies actions $(p_{it}, l_{it}, \chi_{it})$ as functions of the state s_t and the firm's idiosyncratic investment/entry shock v_{it} .
- Value function:

$$V_i(s|\sigma) = \mathbb{E}_v \left[\pi_i(\sigma(s, v), s, v_i) + \beta \int V_i(s'|\sigma) dP(s'|\sigma(s, v), s) \middle| s_t \right]$$

- Markov perfect equilibrium: each σ_i is a best-response to σ_{-i} .

Example: Ericson-Pakes Dynamic Oligopoly, continued

- What does equilibrium look like?
 - State $s_t = (z_t, m_t)$ follows a Markov process: s_t depends (stochastically) on time $t - 1$ states (and actions, which themselves depend stochastically on the time $t - 1$ states).
 - Under certain conditions, there will be a recurrent set of states $R \subset S$. The state s_t will eventually land in R and stay there forever. Moreover, there will be some ergodic (long-run) distribution over R .
 - This ergodic distribution is the long-run (stochastic) steady-state of the industry. This is the small numbers analog to Hopenhayn's steady-state industry equilibrium.
- Applications/Extensions:
 - Benkard: learning by doing in aircraft production.
 - Fershtman and Pakes: collusion (using MPE not SPE!).
 - Doraszelski and Markovich: advertising.
- Stationarity plays an important role. What implications might this have for potential applications?

Computation of EP Equilibrium

- Computation by value iteration (Pakes-McGuire, 1994):
 - ① Guess $\sigma_i^0(s, v)$ and $V_i^0(s)$ for $i = 1, \dots, N$.
 - ② Solve for $\sigma_i^1(s, v)$ and $V_i^1(s)$ for $i = 1, \dots, N$ (given σ^0, V^0)
 - ③ Iterate until convergence.
- Seems just like standard dynamic programming, but difficulties arise:
 - Iterative process is typically *not* a contraction mapping, so if true equilibrium is V^* , V^n may be no closer to V^* than is V^0 . And iterative process may not converge.
 - Often, however, it will converge. If it does, we have an equilibrium. There may, however, be many MPE, which one you find may depend on the starting point!
 - Value iteration becomes more complicated as the state space grows larger, because each iteration requires solving $N \times |S|$ maximization problems (or $|S|$ is one looks only for symmetric MPE as is typical).
 - Here: $|S| = |M| \times |Z|^N$, so dimensionality of the problem grows very fast in Z or N . Typically limits applications to small N , small Z .
 - Some recent work tries various tricks to break the “curse of dimensionality” (e.g. Pakes-McGuire, 2001; Doralzeski and Judd, 

Fitting the Model to Data

- Structural parameters of the model:
 - Profit functions $\pi_1, \dots, \pi_N$
 - Discount factor β
 - Distribution of the private shocks $G_1, \dots, G_N$
- Typically might assume β is known and π_i, G_i are known functions indexed by a finite parameter vector θ : $\pi_i(a, s, v_i; \theta)$ and $G_i(v_i|s; \theta)$.
- Data: actions and state variables over time: $\{a_t, s_t\}_{t=1}^T$.

Estimation: Possible Approaches

- Estimation approach 1: Nested Fixed Point Approach (Rust, 1987)
 - Given a parameter vector θ , compute an equilibrium to the game $V(s, \theta)$ numerically.
 - Use the computed equilibrium to evaluate an objective function based on sample data, e.g. how close is predicted behavior to observed behavior in the data.
 - Nest these steps in a search routine that finds the value of θ that maximizes the objective function.
- Estimation approach 2: Two-Step Approaches (various permutations)
 - Basic idea: substitute (semi- or non-parametric) functions of the data for the continuation values of the game.
 - First stage: estimate value functions from the data (without computing an equilibrium).
 - Second stage: use estimated value functions to estimate parameters θ .

Hotz-Miller: Dynamic Discrete Choice

- Idea: set up dynamic discrete choice problem so that it looks like a standard static discrete choice function, with “choice-specific” value functions taking the place of mean utilities.
- Assume actions $A_i = \{0, 1, \dots, L\}$, and profits

$$\pi_i(a_t, s_t, v_{it}; \theta) = \Pi(a_{it}, s_t; \theta) + v_{it}(a_{it}).$$

- Here θ is the parameters and $v_{it} = \{v_{it}(a)\}_{a \in A_i}$ is a vector of iid choice-specific payoff shocks, with known distribution (e.g. probit, logit).
- Example: entry/exit as discrete choice:
 - Actions $a_i \in \{0, 1\}$ means out/in.
 - State: $s = (m, z_1, \dots, z_N)$, where $z_i \in \{0, 1\}$ signifies if firm is out or in.
 - Profit function:

$$\pi_i(a_t, s_t, v_{it}; \theta) = \begin{cases} \Pi(s_t; \theta) + (1 - a_{it})\phi + v_{it}(a_{it}) & \text{if } z_i = 1 \\ a_{it}\gamma + v_{it}(a_{it}) & \text{if } z_i = 0 \end{cases}$$

where γ is mean entry cost and ϕ is mean scrap value.

- Value function

$$V(s, v) = \max_a \Pi(a, s; \theta) + v(a) + \beta \int V(s', v') dG(v') dP(s'|s, a)$$

- This is a discrete choice problem with mean utilities:

$$v(a, s; \theta) = \Pi(a, s; \theta) + \beta \int V(s', v') dG(v') dP(s'|s, a)$$

- Therefore:

$$\Pr(a|s; \theta) = \Pr(v(a, s; \theta) + v(a) > v(a', s; \theta) + v(a') \text{ for all } a' \neq a)$$

- Logit case:

$$\Pr(a|s; \theta) = \frac{\exp(v(a, s; \theta))}{\sum_{a'} \exp(v(a', s; \theta))}.$$

- With two choices:

$$\begin{aligned} \ln(\Pr(a = 1; s, \theta)) - \ln(\Pr(a = 0; s, \theta)) \\ = v(a = 1, s, \theta) - v(a = 0, s, \theta) \end{aligned}$$

- Given $\Pr(a|s; \theta_0)$ for all a, s , and given knowledge of the distribution of the choice-specific shocks, we can invert to find $v(a, s; \theta_0) - v(a = 0, s; \theta_0)$ for all a, s .

Hotz-Miller Estimation (First Stage)

- Suppose data is generated by the model with “true” parameter θ_0 .
- Estimate $\Pr(a|s; \theta_0)$. Denote this $\hat{\Pr}(a|s)$.
- Next, invert the choice probabilities to find the choice-specific value functions.
- Logit example with two choices:

$$v(a=1, s) - v(a=0, s) = \ln \left(\hat{\Pr}(a=1; s) \right) - \ln \left(\hat{\Pr}(a=0; s) \right)$$

- This is like logit or BLP static demand estimation where we invert the market shares (choice probabilities) to find the mean utilities (choice-specific value functions).
- Next, estimate the “true” value functions:

$$\hat{V}(s, \nu) = \max_a \{ \hat{v}(a, s) + \nu(a) \}.$$

Hotz-Miller Estimation (Second Stage)

- At this point we've obtained direct estimates of the value functions without solving a dynamic programming problem.
- These value functions are a (complicated) function of the true parameters θ_0 . What we want, however, is θ_0 itself.
- Stage 2: use value function estimates to get the parameters.
- Note that for any θ , we can solve the problem

$$\max_a \left\{ \Pi(a, s; \theta) + v(a) + \beta \int \hat{V}(s', v') dG(v') dP(s'|s, a) \right\}.$$

- One proviso is that we need an estimate of $P(s'|s, a)$ to substitute in its place. We get this directly from the data, which is possible if (s, a) is observed each period.
- Solving this problem yields predicted choice probabilities for any θ .
- We then search over different values of θ to find the predicted choice probabilities that are closest to the true choice probabilities according to some distance metric. This is the Hotz-Miller estimate $\hat{\theta}$!

Discussion: Hotz-Miller

- Basic idea: compute value functions “directly” from data without solving a dynamic programming problem. Then estimate profit function parameters.
- Works in decision problems or in games. The main requirement is that it’s possible to do the Hotz-Miller inversion, which works with a fairly general discrete choice set-up, but is easiest with logit/probit structure.
- Less efficient than Rust’s nested fixed point approach. Why? Because we estimate the value functions without imposing any structure that might be implied by the agent’s maximization problem.
- Aguirregabiria-Mira (2002): efficiency gain from iterating. Estimate θ using Hotz-Miller. Solve for optimal policy given $\hat{\theta}$ payoffs to find optimal choice probabilities. Pretend these are the data. Repeat until convergence. Can try this for games too, with stronger assumptions.

Example: Rust Engine Replacement

- One firm.
- State $s_t \in \{1, \dots, M\}$ is machine age/miles.
- Actions $A = \{0, 1\}$ (replace/maintain).
- Private shock $v_t = \{v_t(0), v_t(1)\}$
- Profits at time t :

$$\pi_i(a, s, v; \theta) = \begin{cases} -\mu s + v(0) & \text{if } a = 0 \\ -R + v(1) & \text{if } a = 1 \end{cases}$$

where $\theta = (\mu, R)$ are parameters.

Example: Rust Engine Replacement

- State transitions (deterministic):

$$s_{t+1} = \begin{cases} \min\{s_t + 1, M\} & \text{if } a = 0 \\ 0 & \text{if } a = 1 \end{cases}$$

- Markov policy: $\sigma(s, v)$: whether or not to replace the engine as a function of the engine's age and the idiosyncratic costs of replacement/maintenance.
- Optimal policy will have a cut-off form:

$$\sigma(s, v) = 1 \quad \Leftrightarrow \quad v(1) - v(0) \geq \eta(s).$$

Example: Rust Engine Replacement

- Value function:

$$V(s, v) = \pi(\sigma(s, v), s, v) + \beta \int V(s', v') dG(v') dP(s', v' | \sigma(s, v), s).$$

- Define “choice-specific” value function:

$$v(a = 1, s) = -R + \beta \int V(1, v') dG(v') \quad (1)$$

$$v(a = 0, s) = -\mu s + \beta \int V(\min\{s + 1, M\}, v') dG(v')$$

- Therefore:

$$\Pr(a = 1 | s) = \Pr(v(0) - v(1) \leq v(1, s) - v(0, s))$$

- Assume $v(0), v(1)$ have known distribution, e.g. extreme value or standard normal; in the logit case:

$$\Pr(a = 1 | s) = \frac{\exp(v(1, s))}{\exp(v(1, s)) + \exp(v(0, s))} \quad (2)$$

Example: Rust Engine Replacement

- First-Stage Estimation:

- ① Estimate $\Pr(a = 1|s; \theta_0)$ from the data. Denote this by $\hat{\Pr}(a|s)$
- ② Invert to find $\widehat{v(a = 1, s) - v(a = 0, s)}$ for every s .
- ③ Calculate an estimate of $V(s, v; \theta_0)$. Denote this by $\hat{V}(s, v)$

- Second-Stage Estimation:

- ① For any θ can calculate optimal choice probabilities given continuation values $\hat{V}(s, v)$.
- ② Compare predicted choice probabilities to $\hat{\Pr}(a|s)$.
- ③ Use search routine to make predicted probabilities as close as possible to $\hat{\Pr}(a|s)$.

- BBL propose an alternative two-stage estimator that works for a broader class of dynamic models, including those with discrete and/or continuous choices.
- Recall that given profit functions $\pi_i(a, s, v_i; \theta)$, and transition probabilities $P(s'|s, a)$, the strategy profile $\sigma : S \times \mathcal{V} \rightarrow A$ is an MPE if for every firm i , σ_i is a best response to σ_{-i} . That is, for every firm i , state s and Markov strategy σ'_i

$$V_i(s|\sigma_i, \sigma_{-i}; \theta) \geq V_i(s|\sigma'_i, \sigma_{-i}; \theta)$$

where for any profile σ

$$V_i(s|\sigma; \theta) = \mathbb{E}_v \left[\pi_i(\sigma(s, v), s, v_i; \theta) + \beta \int V_i(s'|\sigma; \theta) dP(s'|s, \sigma(s, v)) \right]$$

- Basic idea in BBL:
 - If we know θ (i.e. all the parameters of the game) and σ_{-i} , we can find player i 's best response $BR(\sigma_{-i}; \theta)$ by solving a dynamic optimization problem.
 - Conversely if we estimate σ_{-i} and σ_i and we know that player i is best-responding to σ_{-i} , we can find the θ that makes $\hat{\sigma}_i$ a best-response to $\hat{\sigma}_{-i}$.
- We do this in a series of steps:
 - 1 Explain *what* firms do in each state (estimate policy functions).
 - 2 Explain *why* they do it (find parameters that rationalize policies).

Bajari, Benkard, Levin: Details

- Consider symmetric Ericsson-Pakes model with policy functions:

$$I(s_t, v_{it}), \chi(s_t, v_{it}) \text{ and } \chi^e(s_t, v_{it}^e).$$

- The entry strategies have the form (exit is similar):

$$\chi_{it} = 1 \text{ if and only if } v_{it}^e \leq \eta(s_t);$$

We must estimate $\eta(s_t)$, easy if we know the distribution of v_{it}^e .

- The investment strategies are $I(s_t, v_{it})$. If investment is monotone in v_{it} , we can invert observed investment to find I . Define:

$$F(x; s) = \Pr(I(s_t, v_{it}) \leq x)$$

so estimating F allows an estimate of I :

$$\hat{I}(s_t, v_{it}) = \hat{F}^{-1}(G(v_{it}); s)$$

where G is the distribution of v_{it} .

- As in HM, also estimate transition probabilities $\hat{P}(s'|a, s)$.

- For *any* player i , strategy $\tilde{\sigma}_i$ and candidate parameter value θ , we can numerically calculate:

$$\hat{V}_i(s; \tilde{\sigma}_i, \hat{\sigma}_{-i}; \theta) = \mathbb{E} \left[\sum_{t=0}^{\infty} \pi_i(\tilde{\sigma}_i(s_t, v_{it}), \hat{\sigma}_{-i}(s_t, v_{-it}), s_t, v_{it}; \theta) \middle| s_0 = s \right]$$

- Easy if profits are linear in θ , so that $\pi_i(a, s, v; \theta) = r_i(a, s, v) \bullet \theta$, which isn't very restrictive, because then:

$$\begin{aligned} \hat{V}_i(s; \tilde{\sigma}_i, \hat{\sigma}_{-i}; \theta) &= \hat{W}_i(s; \tilde{\sigma}_i, \hat{\sigma}_{-i}) \bullet \theta \\ &= \mathbb{E} \left[\sum_{t=0}^{\infty} r_i(\tilde{\sigma}_i(s_t, v_{it}), \hat{\sigma}_{-i}(s_t, v_{-it}), s_t, v_{it}) \middle| s_0 = s \right] \end{aligned}$$

- What we want is some θ for which for all $i, s, \tilde{\sigma}_i$:

$$\hat{V}_i(s; \hat{\sigma}_i, \hat{\sigma}_{-i}; \theta) \geq \hat{V}_i(s; \tilde{\sigma}_i, \hat{\sigma}_{-i}; \theta)$$

in other words, for which the estimated policies are an MPE!

- In practice, may be no such θ , so pick the one that comes “closest”.
- Pick a set of alternative policies $\sigma_i^j, j = 1, \dots, J$, and define the objective function:

$$Q(\theta) = \frac{1}{|S| \times J} \sum_{s, \sigma_i^j} \min\{0, \hat{V}_i(s; \sigma_i^j, \hat{\sigma}_{-i}, \theta) - \hat{V}_i(s; \hat{\sigma}_i, \hat{\sigma}_{-i}; \theta)\}.$$

- The choose θ to maximize this objective function.

Ryan (2009): Portland Cement

- Dynamic competition in Portland Cement industry.
- Data on firm capacity, quantities and prices.
- Formulates EP-type model: estimates demand and marginal costs from static model of quantity competition, and BBL to recover entry and exit costs, and costs of adjusting capacity, i.e. investment.
- Examines possible effects of 1990 CAA environmental regulation.
 - Regulation might raise marginal production costs.
 - Regulation might raise sunk costs of entry.
- Both lead to higher prices, but former is bad for incumbents while latter is good.
- Results: marginal production costs haven't changed much due to regulation, but sunk entry costs have increased substantially.
- Concludes that this has caused a substantial welfare loss for consumers due to decreased entry, but has made incumbents better off as they are more protected from entry.

More Recent Applications

- Holmes (2008): diffusion of Walmart outlets.
- Sweeting (2007) and Jeziorski (2009): format switches after radio station mergers.
- Dunne, Klimek, Roberts and Xu (2009): entry, exit and market structure in service industries.
- Benkard, Bodoh-Creed and Lazarev (2009): long-run effects of airline mergers.
- Collard-Wexler (2008, 2009): demand fluctuations and the long-run effect of mergers in ready-mix concrete industry.